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DLD theory

Assignment #02

1) Given the Boolean function.

$F(A,B,C) = A'B + AB'C + ABC'$

Step 01 :- Convert it into canonical SOP form.

Term 01

$A'B = A'B(C + C')$

$= A'BC + A'BC'$

Term 02

$AB'C$

Term 03

ABC'

So,

$F = A'BC' + A'BC + AB'C + ABC'$

Now write minterms :

Term	Binary	Minterm
$A'BC'$	010	m_2
$A'BC$	011	m_3
$AB'C$	101	m_5
ABC'	110	m_6

Canonical SOP Form

$F(A,B,C) = \Sigma m(2,3,5,6)$

OR

$F = A'BC' + A'BC + AB'C + ABC'$

Step 02.

Canonical POS form

Missing maxterm are:

0, 1, 4, 7

Therefore

$F(A,B,C) = \Pi M(0,1,4,7)$

Now write Max terms:

Maxterm	Binary	POS Term
M6	000	(A+B+C)
M1	001	(A+B+C')
M4	100	(A'+B+C)
M7	111	(A'+B'+C')

canonical POS FORM.

$F = (A+B+C)(A+B+C')(A'+B+C)(A'+B'+C')$

2) Given

$F(A,B,C) = \Sigma m(1,2,3,5,7)$

Step 01 - 3 variable k-MAP.

AB \ C	0	1
00	0	1
01	1	1
11	0	1
10	0	1

Step 02 - Grouping

Group 1 -

cells → 1, 3, 5, 7

common variable :

C

Group 02 -

cell 2, 3

common Variable

$A'B$

Step 03 : Simplified function

$F = C + A'B$

Minimal SOP form

$F = C + A'B$

3) Given Boolean function

$F(A,B,C) = A'Bc + AB' + ABC + A'B'C$

Step 01.

Convert into Minterm.

Term 1:-

$A'Bc = 011 = m3$

Term 02:-

AB'

Expand

$AB'(c + c')$

$= AB'c + AB'c'$

$= m5 + m4$

Term 03

$ABC = 111 = m7$

Term 04

$A'B'c = 001 = m1$

Therefore

$F = \Sigma m(1,3,4,5,7)$

Step 2

3 Variable k-MAP.

AB \ C	0	1
00	0	1
01	0	1
11	0	1
10	1	1

3 variable k MAP.

Step 03 :- Grouping

(Group 1)

cells 1, 3, 5, 7

common variable

c

(Group 2)

cells 4, 5

common variable

AB'

Step 4 : Simplified function

$F = C + AB'$

Minimal SOP FORM.

$$F = C + AB'$$